Paul Rax

PhD student at INRIA Paris/ DIENS

Équipe Mathnet, INRIA Paris
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Education

2024 – **PhD thesis**, *MATHNET team(INRIA Paris) and DI ENS (ENS-PSL)*, Under the supervision of F.Baccelli and R.Lachièze-Rey

2024 **Internship**, *"Projet Long de Recherche" (fourth year at ENS de Lyon)*, UIBK (Innsbruck)

2023-2024 **Internship**, *"Projet Long de Recherche"*, LPSM (Sorbonne Université)

2022-2023 **Gap year**, *"césure de diplôme"*

2021-2022 **Master 2 Mathematics of Randomness**, *Université Paris-Saclay*, Probability and Stat specialization

2020-2021 **Master 1 Advanced Maths**, *ENS de Lyon*

2019-2020 **Bachelor in Fundamental Math**, *ENS de Lyon*

2019-2024 **Student at ENS de Lyon**, *ENS de Lyon*, Math Department

2017-2019 **MPSI-MP* (Classe préparatoire)**, *Lycée Blaise Pascal*, Orsay

Diplomas

2025 **ENS de Lyon Diploma**, *Mathematics*, ENS de Lyon

2022 **Master Mathematics and Applications**, *M2 Mathematics of Randomness*, Université Paris-Saclay

2020 **Bachelor in Fundamental Mathematics**, *ENS de Lyon*

2019 **Admitted at ENS de Lyon**, *Computer science exam*

Teaching activities

2025 **Exercise sessions**, *Last year Bachelor students*, Advanced Probabilities, Sorbonne université

2021-2022 **"Colleur" (oral examiner) in Math**, *PC*, Lycée Blaise Pascal, Orsay*, 2h/week

2020-2021 **Colleur in Math**, *MPSI 831, Lycée du Parc, Lyon*, 2h/week

2020 **Colleur in Math**, *MPSI A, Lycée Blaise Pascal, Orsay*, 2h/week
Volunteering during lockdown (april-june)

2019-2020 **Volunteering homework help**, *ENSeigner association*

2017-2018 **Jury for $TFJM^2$ (french version of ITYM)**, *Île-de-France regional tournament*

Research

2026 **The Relay Random Tree : A Stochastic Geometry Approach of Multihop Relay in an Urban Visibility Setting**, *Joint work with François Baccelli*, arXiv :2605.03851

2025 **Stochastic domination and lifts of random variables in percolation theory**, *Joint work with Sébastien Martineau and Rémy Poudevigne*, arXiv :2504.02427, Annales Henri Lebesgue

Administrative duties

2026 **Secretary for the entrance exams at ENS– PSL**, *Concours CPGE – MP/MPI/MI*

2025 – **Co-organizer for the MATHNET team seminar**

Language skills

English **Fluent**, *official certification only done for C1*, Cambridge Advanced level of English

Allemand **B1**, *Deutsches Sprachdiplom (DSD)*